Week 4 Make controls reliable

Step by step · GROW computing · 40 minutes

Name: ____________________ Date: _________

I can create and test all four arrow controls in my maze.

Point, say, draw or write. Someone may record your exact words.

Arrival > Starter > I Do > We Do
I Do 2 > We Do 2 · Task > Independent Task > Exit

Arrival 0 to 3 minutes

Open your own W03_Maze.sb3.

Starter 3 to 6 minutes

The up key changes x by 10. Predict the direction Player moves.

My prediction

____

I Do 6 to 11 minutes

Watch the model. Point to the sprite or block that matters.

We Do 11 to 18 minutes

Predict. Ask a partner or adult to check your idea. Test together. Explain what changed.

Shared test	We expect	We see
Reset, then up	Moves up	
Reset, then right/up/left/down	Returns to start	

I Do 2 18 to 21 minutes

Up and down use y. Up adds 10. Down adds -10.

We Do 2 · Task 21 to 25 minutes

Show your next action. Choose a task route with your teacher.

Reset Player. Predict right, up, left, down.

My coding task

Step by step · Independent Task 25 to 35 minutes

☐ Select Player.

☐ Place it clear of every wall. Write or point to its x and y.

☐ Add when green flag clicked. Join go to x and y with those numbers.

☐ Build right arrow with change x by 10. Test it.

☐ Build left arrow with change x by -10. Test it.

☐ Build up arrow with change y by 10. Test it.

☐ Build down arrow with change y by -10. Test it.

☐ Reset. Test right, up, left, down.

My safe start: x ______ y ______

One test I ran	Expected result	Actual result

Exit 35 to 40 minutes

☐ Save as W04_Controls.sb3.
☐ Load the saved file.
☐ Run the check or select the three sprites.

File > Save to your computer. Then File > Load from your computer. Keep the .sb3 file in your school folder.

Help for the next step

Step by step reference · Keep beside your device

Find this	Use it like this
Right arrow	Events: when right arrow key pressed. Motion: change x by 10.
Left arrow	Events: when left arrow key pressed. Motion: change x by -10.
Up arrow	Events: when up arrow key pressed. Motion: change y by 10.
Down arrow	Events: when down arrow key pressed. Motion: change y by -10.
Green flag	Events: when green flag clicked. Motion: go to x [start x] y [start y].

A paper controls check

Model start: x = -180, y = -120.
One key press moves one square on this grid.

y / x	-190	-180	-170
-110			
-120		START	
-130			

Give a partner one key input. Move a token.
Reset the token before a new test.

My help signals

NOW	NEXT	HELP
Show one action.	Show the next block.	Read or model it.
PAUSE: I need a break.	RETURN: same action.	CHECK: watch my test.

Paper rehearsal prepares the idea. Complete the code, tests and save on a device.

My voice

Tell someone your idea. Ask what changed.
“What I said, and what it changed”

My idea and the reply